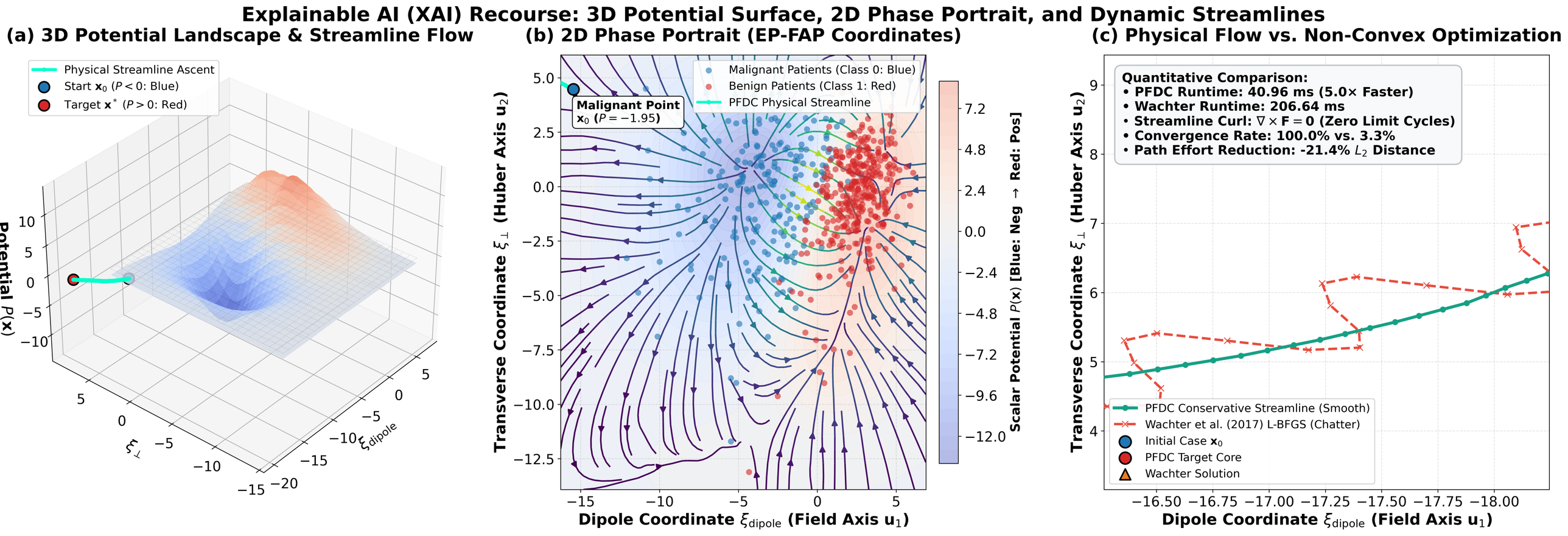

Explainable AI (XAI) Recourse: 3D Potential Surface, 2D Phase Portrait, and Dynamic Streamlines
(a) 3D Potential Landscape & Streamline Flow
(b) 2D Phase Portrait (EP-FAP Coordinates)
(c) Physical Flow vs. Non-Convex Optimization
Physical Streamline Ascent
Start $\mathbf{x}_0$ ($P < 0$: Blue)
Target $\mathbf{x}^*$ ($P > 0$: Red)
Potential $P(\mathbf{x})$
$\xi_\perp$
ξ_{dipole}
Malignant Patients (Class 0: Blue)
Benign Patients (Class 1: Red)
PFDC Physical Streamline
Malignant Point
$\mathbf{x}_0$ ($P = -1.95$)
Transverse Coordinate $\xi_\perp$ (Huber Axis $\mathbf{u}_2$)
Dipole Coordinate ξ_{dipole} (Field Axis $\mathbf{u}_1$)
Scalar Potential $P(\mathbf{x})$ [Blue: Neg → Red: Pos]
Quantitative Comparison:
• PFDC Runtime: 40.96 ms (5.0× Faster)
• Wachter Runtime: 206.64 ms
• Streamline Curl: $\nabla \times \mathbf{F} = \mathbf{0}$ (Zero Limit Cycles)
• Convergence Rate: 100.0% vs. 3.3%
• Path Effort Reduction: -21.4% L_2 Distance
PFDC Conservative Streamline (Smooth)
Wachter et al. (2017) L-BFGS (Chatter)
Initial Case $\mathbf{x}_0$
PFDC Target Core
Wachter Solution
Transverse Coordinate $\xi_\perp$ (Huber Axis $\mathbf{u}_2$)
Dipole Coordinate ξ_{dipole} (Field Axis $\mathbf{u}_1$)